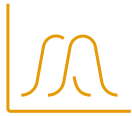


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PROBABILITY & STATISTICAL DATA ANALYSIS

CHAPTER 5.2 REVIEW

HYPOTHESIS TESTING



**Formulate
Hypothesis
Statement**

Clearly define null
and alternative
hypotheses



**Calculate Test
Statistic**

Summarize data
numerically in
relation to
hypothesis



**Establish
Significance
Level**

Determine threshold
for rejecting null
hypothesis



**Conclusion /
Decision Rule**

Interpretation
of results and
decision-making

STEP 1: Hypothesis statement

Null Hypothesis: H_0

- The **null hypothesis** (denoted by H_0) is a statement that the value of a population parameter (such as proportion, mean, or standard deviation) is **equal to** some claimed value.
- We test the null hypothesis directly either reject H_0 or fail to reject H_0 .
- The symbolic form of the null hypothesis is:

=

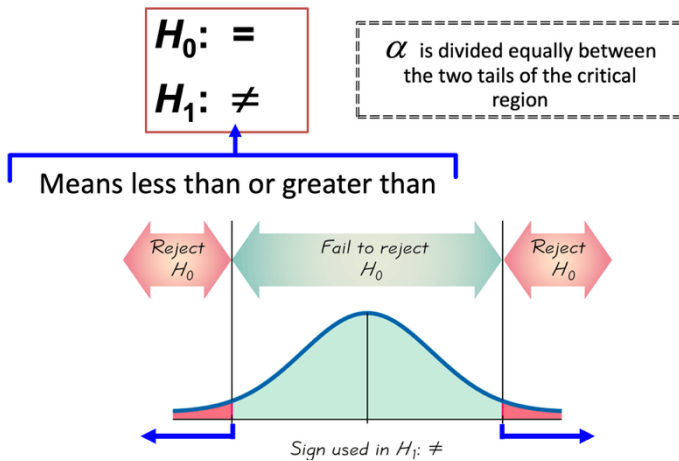
Alternative Hypothesis: H_1

- The **alternative hypothesis** (denoted by H_1 or H_a or H_A) is the statement that the parameter has a value that somehow differs from the null hypothesis.
- The symbolic form of the alternative hypothesis must use one of these symbols:

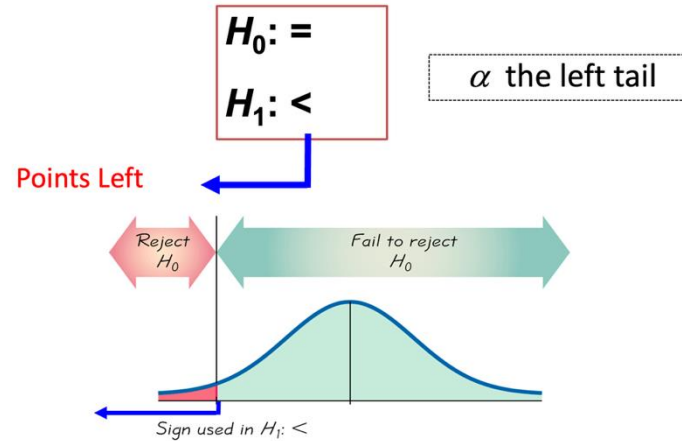
$\neq, <, >.$

STEP 1: Hypothesis statement

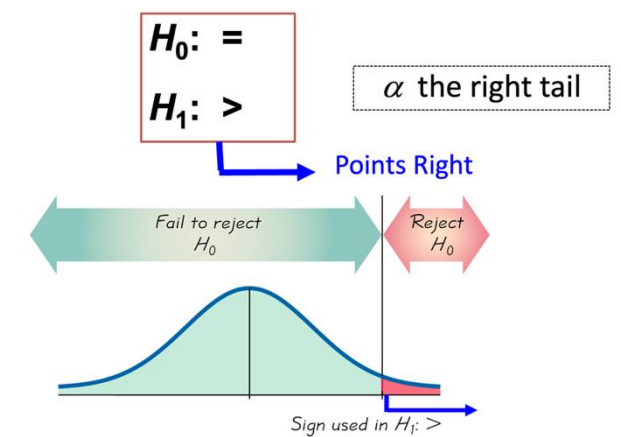
Two-tailed Test



Left-tailed Test



Right-tailed Test



STEP 2: Test Statistic

- A **test statistic** is computed using sample data and is the value used to reach a conclusion to reject or fail to reject H_0 .
- There are term/formula to calculate test statistic:

TEST STATISTIC FOR PROPORTION (ρ)

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

TEST STATISTICS FOR MEAN (μ)

variance known
or $n > 30$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

variance unknown
and $n \leq 30$

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

TEST STATISTICS FOR VARIANCE (σ^2) / STANDARD DEVIATION (σ)

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

STEP 3: Significance Level & Level of Confidence

- The **significance level** (denoted by α) is the probability that the test statistic will fall in the critical region when the null hypothesis is actually true.
- Common choices for α are 0.05, 0.01, and 0.10.

Aspect	Confidence Level (How confident are we?)	Significance Level (α) How much risk of error are we willing to accept?
Meaning	The level of confidence that the result is correct	The probability of making a wrong decision by rejecting a true null hypothesis
Focus	Confidence in the estimate or result	Risk of error in hypothesis testing
Common Values	90%, 95%, 99%	0.10, 0.05, 0.01
Relationship	Confidence Level = $1 - \alpha$	$\alpha = 1 - \text{Confidence Level}$
Usage	Confidence intervals	Hypothesis testing

STEP 4: Conclusion/Decision rule

A lecturer wants to know whether students study **more than 5 hours per week** on average, with significance level: $\alpha = 0.05$. Assume the test statistics is $z = 1.80$.

$$H_0: \mu = 5$$

$$H_1: \mu > 5$$

P-value

REJECT H_0

$$P\text{-value} \leq \alpha$$

FAIL TO REJECT H_0

$$P\text{-value} > \alpha$$

This is a right-tailed test, so $(P(Z > 1.8))$: 0.0359.

Because $0.0359 < \alpha$ (0.05), we **REJECT** the H_0 . Thus, there is enough evidence to conclude that students study more than 5 hours per week

Critical Region

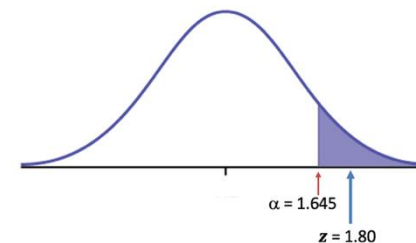
REJECT H_0

The **test statistic falls** within the critical region

FAIL TO REJECT H_0

The **test statistic does not fall** within the critical region

This is a right-tailed test, the critical value for α (0.05) = 1.645.



Because z (1.80) $> \alpha_{0.05}$ (1.645) and it falls in the critical region, we **REJECT** the H_0 .

Thus, there is enough evidence to conclude that students study more than 5 hours per week

P-value

Critical Region

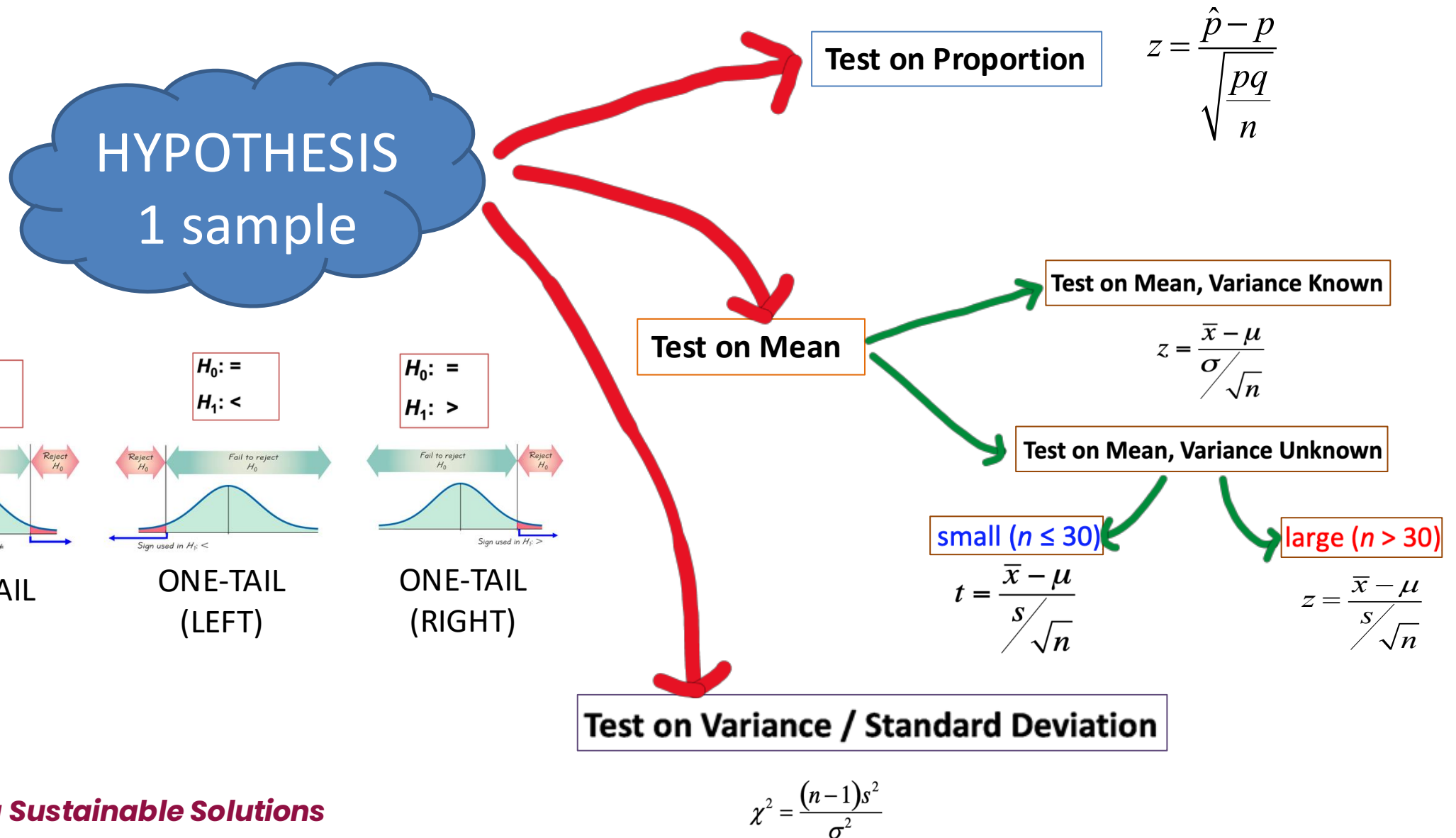
You can **use either one (avoid using both):**

- will lead to the same conclusion if applied correctly
- they are just **different ways of expressing statistical significance.**

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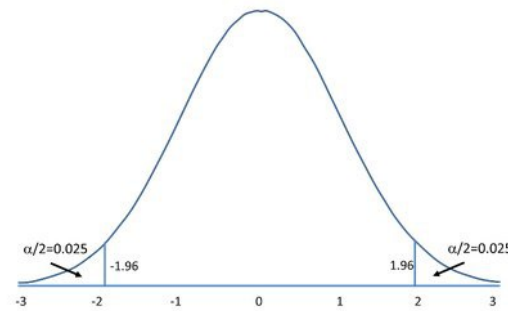
CHAPTER 5 SUMMARY



STEP 4: Conclusion/Decision rule

REJECTION CRITERION FOR Z-TEST

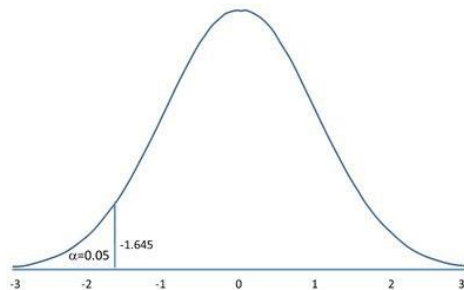
TWO-TAILED TEST



Rejection Region for Two-Tailed Z Test ($H_1: \mu \neq \mu_0$) with $\alpha = 0.05$
 The decision rule is: Reject H_0 if $Z \leq -1.960$ or if $Z \geq 1.960$.

Two-Tailed Test	
α	Z
0.20	1.282
0.10	1.645
0.05	1.960
0.010	2.576
0.001	3.291
0.0001	3.819

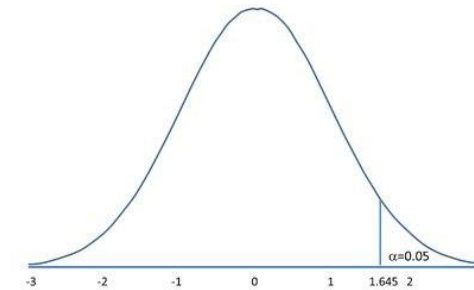
LEFT-TAILED TEST



Rejection Region for Lower-Tailed Z Test ($H_1: \mu < \mu_0$) with $\alpha = 0.05$
 The decision rule is: Reject H_0 if $Z \leq -1.645$.

Lower-Tailed Test	
α	Z
0.10	-1.282
0.05	-1.645
0.025	-1.960
0.010	-2.326
0.005	-2.576
0.001	-3.090
0.0001	-3.719

RIGHT-TAILED TEST



Rejection Region for Upper-Tailed Z Test ($H_1: \mu > \mu_0$) with $\alpha = 0.05$
 The decision rule is: Reject H_0 if $Z \geq 1.645$.

Upper-Tailed Test	
α	Z
0.10	1.282
0.05	1.645
0.025	1.960
0.010	2.326
0.005	2.576
0.001	3.090
0.0001	3.719

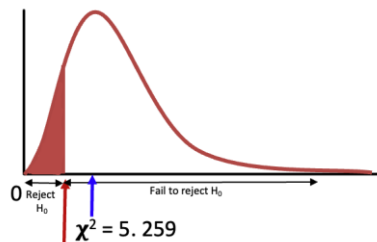
REJECTION CRITERION FOR χ^2 TEST

$$H_0 : \sigma = 3.7 \text{ min}$$

$$H_1 : \sigma < 3.7 \text{ min, with } \alpha = 0.05$$

$$\chi^2 = 5.259$$

$$\chi^2_{0.95,8} = 2.733$$



$$\chi^2_{0.95,8} = 2.733$$

We see that $\chi^2 (5.259) > \chi^2_{\alpha} (2.733)$, it does not fall in the critical region; we **FAIL TO REJECT** the H_0 .

Thus, there is enough evidence to support that the standard deviation in length of response times is 3.7 minutes.

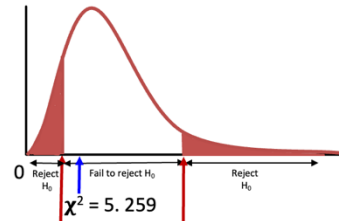
$$H_0 : \sigma = 3.7 \text{ min}$$

$$H_1 : \sigma \neq 3.7 \text{ min, with } \alpha = 0.05$$

$$\chi^2 = 5.259$$

$$\alpha/2 = 0.025 \text{ and } (1-\alpha/2) = 0.975$$

$$\chi^2_{0.025,8} = 2.180 \text{ and } \chi^2_{0.975,8} = 17.535$$



$$\chi^2_{0.975,8} = 2.180$$

$$\chi^2_{0.025,8} = 17.535$$

We see that

$\chi^2_{\alpha} (2.180) < \chi^2 (5.259) < \chi^2_{\alpha} (17.535)$, it does not fall in the critical region; we **FAIL TO REJECT** the H_0 .

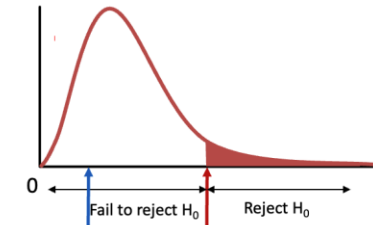
Thus, there is enough evidence to support that the standard deviation in length of response times is 3.7 minutes.

$$H_0 : \sigma = 3.7 \text{ min}$$

$$H_1 : \sigma > 3.7 \text{ min, with } \alpha = 0.05$$

$$\chi^2 = 5.259$$

$$\chi^2_{0.95,8} = 15.507$$



$$\chi^2 = 5.259$$

$$\chi^2_{0.05,8} = 15.507$$

We see that $\chi^2 (5.259) < \chi^2_{\alpha} (15.507)$, it does not fall in the critical region; we **FAIL TO REJECT** the H_0 .

Thus, there is enough evidence to support that the standard deviation in length of response times is 3.7 minutes.